API documentation

Base URL for local development: http://localhost:8000. Interactive OpenAPI docs are at /docs; the schema is at /openapi.json. All request bodies and JSON responses use application/json unless noted. Examples below are representative; model values and dataset status vary by runtime.

Shared property request format

PropertyInput fields are optional. Supply a natural-language query, structured values, or both. A valuation still requires at least one recognizable property detail. Omitted input values may be extracted from the query or imputed using training medians.

Field	JSON type and constraints
query	string, max 1,200 characters, default ""; instruction-like text is rejected
neighborhood	string, max 60, valid place-name characters
OverallQual, OverallCond	integer, 1-10
GrLivArea	number, 100-20,000
TotalBsmtSF	number, 0-20,000
GarageCars	number, 0-10
GarageArea	number, 0-5,000
YearBuilt, YearRemodAdd	integer, 1800-2026
GarageYrBlt	number, 1800-2026
FullBath	number, 0-10
BedroomAbvGr	number, 0-20
TotRmsAbvGrd	number, 0-30
LotArea	number, 0-200,000

Shared example:

```
{"query":"3 bedroom home in NAmes, 1800 sq ft, built 2005, quality 7"}
```

Endpoint index

Method	Endpoint	Request	Response
GET	/api/health	None	Runtime status, persistence availability, dataset/model/retrieval/workflow status.
GET	/api/config/status	None	LLM configuration status/model names and runtime status; never returns keys.
GET	/api/model-info	None	Active feature names and runtime status.
POST	/api/valuate	PropertyInput	Full estimate, range, comparables, evidence, impacts, safeguards, judge result, and DeepEval pending state.
POST	/api/search	PropertyInput	Parsed features, comparable properties, retrieval metadata, status.
POST	/api/compare	PropertyInput	Same response as /api/search.
POST	/api/what-if	WhatIfIn below	Original and revised estimates, difference, changed fields, caveat.
POST	/api/feedback	FeedbackIn below	{"status":"saved"} or HTTP 503 if persistence fails.
POST	/api/evaluate	PropertyInput	Optional explanation judge object; does not affect estimate.
POST	/api/deepeval/evaluate	PropertyInput	The single GEval evaluation object or unavailable result.

GET endpoints

‘GET /api/health‘

Request: no body.

Response format: status (normally "ok"), persistenceavailable boolean, plus datasetloaded, modelloaded, chromaloaded, langgraphenabled, propertycount, and datasetpath.

Sample response (paths and component availability depend on the machine):


```
{"status":"ok","persistence_available":true,"dataset_loaded":true,"model_loaded":true,
"chroma_loaded":true,"langgraph_enabled":true,"property_count":2930,"dataset_path":"da
ta/AmesHousing.csv"}
```


‘GET /api/config/status‘

Request: no body. Response includes externalllmconfigured boolean, llmprovider, configured model names or null/empty values, and the same runtime status fields as /api/health. Secret values are never returned.


```
{"external_llm_configured":false,"llm_provider":"anthropic","primary_llm_model":"claud
e-sonnet-4-6","fallback_llm_model":null,"dataset_loaded":true,"model_loaded":true,"chr
oma_loaded":true,"langgraph_enabled":true,"property_count":2930,"dataset_path":"data/A
mesHousing.csv"}
```


‘GET /api/model-info‘

Request: no body. Response contains features (array of active model feature names) and runtime status fields.


```
{"features":["OverallQual","OverallCond","GrLivArea","TotalBsmtSF","GarageCars","Garag
eArea","YearBuilt","YearRemodAdd","FullBath","BedroomAbvGr","TotRmsAbvGrd","LotArea"],
"dataset_loaded":true,"model_loaded":true,"chroma_loaded":true,"langgraph_enabled":tru
e,"property_count":2930,"dataset_path":"data/AmesHousing.csv"}
```


POST endpoints

‘POST /api/valuate‘

Request: shared PropertyInput JSON.

Success response format includes status, propertyfeatures, predictedvalue, valuationrange (lower, upper), confidence, risklevel, riskfactors, valuationmethod, fallbackused, fallbackreason, comparables (array), featureimpacts (array), retrieval, evidence (array), explanation, evaluation (LLM judge/deterministic result), deepevalevaluation (initially pending), and queryid (integer or null).


```
{
  "status":"success",
  "property_features":{"neighborhood":"North_Ames","BedroomAbvGr":3,"GrLivArea":1800,"
YearBuilt":2005,"OverallQual":7},
  "predicted_value":208313.0,
  "valuation_range":{"lower":187202.0,"upper":229425.0},
  "confidence":0.71,
  "risk_level":"MEDIUM",
```

```
  "risk_factors":[],
  "valuation_method":"model_with_comparables",
  "fallback_used":false,
  "fallback_reason":null,
  "comparables":[{"property_id":42,"SalePrice":205000,"Neighborhood":"North_Ames","sim
ilarity_score":81.2,"document":"Property in North_Ames neighborhood; ..."}],
  "feature_impacts":[{"feature":"OverallQual","impact":12000.0,"direction":"positive",
"method":"shap_tree_explainer"}],
  "retrieval":{"retrieval_mode":"chroma_bm25_rrf","fallback_used":false,"lexical_metho
d":"BM25Okapi","semantic_store":"ChromaDB"},
  "evidence":["Property in North_Ames neighborhood; ..."],
  "explanation":"Evidence-based estimate from historical comparable properties. This
is an estimate, not a guarantee.",
  "evaluation":{"groundedness":1.0,"relevance":1.0,"completeness":1.0,"evidence_usage"
:1.0,"reasoning_quality":null,"verdict":"Deterministic validation only; no LLM
judgment was produced.","unsupported_claims":[],"judge_method":"deterministic_judge_fa
llback","fallback_used":true},
  "deepeval_evaluation":{"status":"pending","metric_name":"Property Recommendation
Quality","score":null,"verdict":"Evaluation pending","reason":"DeepEval will run
separately so it cannot delay the valuation."},
  "query_id":12
}
```


The example is illustrative, not a guaranteed output. If valuation is unavailable because the dataset is missing, response status is setuprequired with message and parsed propertyfeatures.

‘POST /api/search‘

Request: shared PropertyInput JSON. Response format: status, propertyfeatures, comparables, retrieval, and optional message.


```
{"query":"3 bedroom home in NAmes, 1800 sq ft, built 2005"}
```



```
{"status":"success","property_features":{"neighborhood":"North_Ames","BedroomAbvGr":3,
"GrLivArea":1800,"YearBuilt":2005},"comparables":[{"property_id":42,"SalePrice":205000
,"Neighborhood":"North_Ames","similarity_score":81.2,"semantic_score":0.73,"bm25_score
":2.1,"document":"Property in North_Ames neighborhood; ..."}],"retrieval":{"exact_matc
h_count":18,"relaxed_match_count":0,"retrieval_mode":"chroma_bm25_rrf","fallback_used"
:false,"filters_applied":["neighborhood","YearBuilt","BedroomAbvGr","GrLivArea"],"sema
ntic_available":true,"lexical_method":"BM25Okapi","semantic_store":"ChromaDB"}}
```


‘POST /api/compare‘

Request and response formats are identical to /api/search; this endpoint delegates to the same comparable search operation.


```
{"query":"4 bedroom, 2 bath house in Old Town with 1600 sq ft"}
```



```
{"status":"success","property_features":{"neighborhood":"Old_Town","BedroomAbvGr":4,"F
ullBath":2,"GrLivArea":1600},"comparables":[{"property_id":58,"SalePrice":184000,"Neig
hborhood":"Old_Town","similarity_score":77.4,"semantic_score":0.69,"bm25_score":1.8,"d
ocument":"Property in Old_Town neighborhood; ..."}],"retrieval":{"exact_match_count":9
,"relaxed_match_count":0,"retrieval_mode":"chroma_bm25_rrf","fallback_used":false,"fil
ters_applied":["neighborhood","BedroomAbvGr","FullBath","GrLivArea"],"semantic_availab
le":true,"lexical_method":"BM25Okapi","semantic_store":"ChromaDB"}}
```

‘POST /api/what-if‘

Request format is an object containing a nested property (PropertyInput) and a changes object. Allowed changes are the property fields other than query, and each value is validated against its normal bounds.

```
{"property":{"query":"3 bedroom home in
NAmes","OverallQual":6},"changes":{"OverallQual":8,"GrLivArea":1900}}
```

Success response format: status, originalprediction, newprediction, difference, changedfeatures, and a non-causal note.

```
{"status":"success","original_prediction":198500.0,"new_prediction":224100.0,"differen
ce":25600.0,"changed_features":{"OverallQual":8,"GrLivArea":1900},"note":"Model-based
scenario estimate; it is not a causal effect."}
```

If the original property cannot be valued, the underlying valuation result is returned. Invalid or unsupported change fields yield HTTP 422.

‘POST /api/feedback‘

Request format (FeedbackIn): optional integer/null queryid, required integer rating in -1..1, optional string comment up to 1,000 characters.

```
{"query_id":12,"rating":1,"comment":"The comparable homes were relevant."}
```

```
{"status":"saved"}
```

If SQLite persistence is unavailable, response is HTTP 503 with a detail message. Feedback cannot change the numeric estimate; repeated votes can add a small comparable-ranking prior.

‘POST /api/evaluate‘

Request: shared PropertyInput JSON. On successful valuation response format is status, evaluation, and note. Evaluation contains five score fields (groundedness, relevance, completeness, evidenceusage, reasoningquality), plus verdict, unsupportedclaims, judge method/model, and fallback fields. Scores may be null when a deterministic check cannot produce an LLM-only measure.

```
{"query":"3 bedroom home in NAmes, 1800 sq ft, built 2005"}
```

```
{"status":"success","evaluation":{"groundedness":0.9,"relevance":0.9,"completeness":0.
8,"evidence_usage":0.9,"reasoning_quality":0.8,"verdict":"Explanation is grounded in
supplied evidence.","unsupported_claims":[],"judge_method":"llm_as_judge","judge_model
":"claude-sonnet-4-6","fallback_used":false},"note":"LLM-as-Judge evaluates
explanation quality only; it never modifies the numeric valuation."}
```

When valuation is not successful, response has evaluation: null and the valuation status/message.

‘POST /api/deepeval/evaluate‘

Request: shared PropertyInput JSON. Runs the single GEval metric Property Recommendation Quality (threshold 0.6); no other DeepEval metrics are implemented. Response format is outer

status plus evaluation with status, metricname, score (0..1 or null), verdict, and reason.

```
{"query":"3 bedroom home in NAmes, 1800 sq ft, built 2005"}
```

```
{"status":"success","evaluation":{"status":"success","metric_name":"Property
Recommendation Quality","score":0.82,"verdict":"Meets quality threshold","reason":"The
recommendation uses the property details and is grounded in the comparable
evidence."}}
```

If DeepEval/API evaluation fails or a successful valuation is unavailable, the route returns an unavailable evaluation with score: null, verdict: "Evaluation unavailable", and a reason; valuation itself remains available.

Common errors and availability

- 422 Unprocessable Entity: invalid JSON/schema values, property bounds, unsafe instruction-like query, or unsupported what-if fields. FastAPI returns validation details.
- 503 Service Unavailable: feedback could not be persisted. Valuation remains available.
- Optional LLM and DeepEval failures are represented by evaluation fallback/unavailable fields and do not replace or break numeric valuation.
- CORS permits the local Vite origin http://localhost:5173; deployment origins must be configured in backend/app/main.py.